

# ZIYAN WANG

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Google Scholar ◊ Github ◊ LinkedIn

## EDUCATION

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### King's College London (KCL)

*Ph.D. in Computer Science*

Dec. 2022 - Now

London, UK

- Advisors: Prof. Yali Du, Prof. Sanjay Modgil

### Carnegie Mellon University (CMU)

*Visiting Ph.D. Student*

Feb. 2025 - Jun. 2025

Pittsburgh, US

- Advisor: Prof. Fei Fang

### University College London (UCL)

*M.S. in Robotics and Computation*

Sep. 2020 - Dec. 2021

London, UK

- Graduated with **Distinction** Rank: top 15%
- Advisors: Prof. Jun Wang

### Eötvös Loránd University (ELTE)

*B.S. in Computer Science*

Sep. 2016 - Jun. 2019

Budapest, Hungary

- GPA: 4.73/5.0 Rank: 3/78

## RESEARCH INTERESTS

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My research interests lie in multi-agent reinforcement learning, LLM-based agents, safe and aligned autonomous systems, and AI for social impact. I am particularly interested in developing algorithms for agent collaboration in complex environments, enabling agents to understand and follow natural-language constraints, studying multi-agent LLM communication, and evaluating safety failures in open-ended agentic systems.

## WORKING EXPERIENCE

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### Microsoft Research - Research Intern

*Supervisor: Kirill Kalinin*

May 2026 - Present

Cambridge, UK

- Research on multi-agent LLM communication, coordination, and collaborative agent behavior.

### University of Oxford - IDAI Fellowship

*Supervisors: Dr. Adel Bibi, Prof. Philip Torr*

Feb 2026 - Now

Oxford, UK

- Focus on Real-Time Multi-Agent LLMs Anomaly Detection and Monitoring

### Microsoft Research - Research Intern - AI Frontier Group

*Supervisors: Vaishnavi Shrivastava, Prof. Dimitris Papailiopoulos*

Sep 2025 - Dec 2025

Redmond, US

- Focus on LLM Pre-training and Reasoning

## PUBLICATIONS

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\* denotes equal contribution.

## Refereed Conference/Journal Publications

1. M<sup>3</sup>HF: Multi-agent Reinforcement Learning from Multi-phase Human Feedback of Mixed Quality  
**Ziyan Wang**, Zhicheng Zhang, Fei Fang, Yali Du  
*The Forty-Second International Conference on Machine Learning (ICML 2025)*
2. SMALL: Safe Multi-agent Reinforcement Learning with Natural Language Constraints  
**Ziyan Wang**, Meng Fang, Tristan Tomilin, Fei Fang, and Yali Du  
*Alignment Track of the 40th Annual AAAI Conference on Artificial Intelligence (AAAI 2026)*
3. MACCA: Offline Multi-agent Reinforcement Learning with Causal Credit Assignment  
**Ziyan Wang**, Yali Du, Yudi Zhang, Meng Fang, and Biwei Huang  
*The Transactions on Machine Learning Research (TMLR)*
4. Policy Learning from Tutorial Books via Understanding, Rehearsing and Introspecting  
Xiong-Hui Chen\*, **Ziyan Wang\***, Yali Du, *et al.*  
*The Conference on Neural Information Processing Systems (NeurIPS 2024 **Oral**)*
5. Learning to Discuss Strategically: A Case Study on One Night Ultimate Werewolf  
Xuanfa Jin\*, **Ziyan Wang\***, Yali Du, *et al.*  
*The Conference on Neural Information Processing Systems (NeurIPS 2024)*
6. Chessgpt: Bridging policy learning and language modeling  
Xidong Feng, Yicheng Luo, **Ziyan Wang**, *et al.*  
*The Conference on Neural Information Processing Systems (NeurIPS 2023)*
7. Interpretable Reward Redistribution in Reinforcement Learning: A Causal Approach  
Yudi Zhang, Yali Du, Biwei Huang, **Ziyan Wang**, *et al.*  
*The Conference on Neural Information Processing Systems (NeurIPS 2023)*
8. Sauté RL: Almost Surely Safe Reinforcement Learning Using State Augmentation  
Aivar Sootla, Alexander I. Cowen-Rivers, Taher Jafferjee, **Ziyan Wang**, *et al.*  
*The International Conference on Machine Learning (ICML 2022 **Spotlight**)*
9. Bridging Confidence and Competence: Evaluating Self-Assessment Alignment in LLM Mathematical Reasoning  
Mingze Zhong, Zijing Shi, **Ziyan Wang**, *et al.*  
*The Pacific Rim International Conference on Artificial Intelligence (PRICAI 2025)*
10. Safe Reinforcement Learning with Free-form Natural Language Constraints and Pre-Trained Language Models  
Xingzhou Lou, Junge Zhang, **Ziyan Wang**, *et al.*  
*The International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS 2024)*
11. Evaluating Generalization Capabilities of LLM-Based Agents in Mixed-Motive Scenarios Using Concordia  
Chandler Smith, Marwa Abdulhai, Manfred Diaz, . . . , **Ziyan Wang**, *et al.* (Concordia Collaboration; 70+ co-authors)  
*The Conference on Neural Information Processing Systems (NeurIPS 2025)*

## Preprints

1. BazaarBench: Open-Ended Multi-Agent Safety Evaluation in C2C Marketplaces  
**Ziyan Wang**, Shuqing Shi, James Oldfield, Samuele Marro, Jialin Yu, Francesco Pinto, Philip

Torr, Yali Du, Adel Bibi  
*Under Review*

2. Memento: Teaching LLMs to Manage Their Own Context  
Vasilis Kontonis, Yuchen Zeng, Shivam Garg, Lingjiao Chen, Hao Tang, **Ziyan Wang**, Ahmed Awadallah, Eric Horvitz, John Langford, Dimitris Papailiopoulos  
*Under Review*
3. VAM: Verbalized Action Masking for Controllable Exploration in RL Post-Training—A Chess Case Study  
Zhicheng Zhang\*, **Ziyan Wang\***, Yali Du, Fei Fang  
*Under Review*
4. Learning Instruction-Following Policies through Open-Ended Instruction Relabeling with Large Language Models  
Zhicheng Zhang\*, **Ziyan Wang\***, Yali Du, Fei Fang  
*Under Review*
5. Multi-agent Constrained Policy Optimisation  
Shangding Gu, Jakub Grudzien Kuba, Muning Wen, Ruiqing Chen, **Ziyan Wang**, *et al.*  
*Under Review*

## ACHIEVEMENTS

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|--------------------------------|-----------------------------|
| Oxford IDAI Fellowship         | <i>Feb. 2026</i>            |
| NeurIPS 2024 Scholar Award     | <i>Oct. 2024</i>            |
| NeurIPS 2024 Oral Presentation | <i>Sep. 2024</i>            |
| King's PhD Scholarship         | <i>Dec. 2022 - Dec.2026</i> |
| Hungarian State Scholarship    | <i>Sep. 2016 - Jun.2019</i> |

## TEACHING

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|-------------------------------------------------------------------------------|--------------------|
| Teaching, Oxford Machine Learning Summer School - DNN + Optimisation          | <i>Summer 2024</i> |
| Teaching Assistant, Oxford Machine Learning Summer School (ML x Fundamentals) | <i>Summer 2023</i> |
| Teaching Assistant, Optimisation Methods (Level 6), King's College London     | <i>Spring 2023</i> |

## ACADEMIC SERVICE

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### Conference Reviewer

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|------------------------------------------------------------------------------|----------------------------|
| International Conference on Machine Learning (ICML)                          | <i>2023/2024/2025/2026</i> |
| Conference on Neural Information Processing Systems (NeurIPS)                | <i>2023/2024/2025/2026</i> |
| International Conference on Learning Representations (ICLR)                  | <i>2024/2025/2026</i>      |
| International Conference on Artificial Intelligence and Statistics (AISTATS) | <i>2025/2026</i>           |
| International Conference on Autonomous Agents and Multiagent Systems (AAMAS) | <i>2025/2026</i>           |

### Journal Reviewer

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|------------------------------------------------------------|--------------|
| IEEE Robotics and Automation Letters (RAL)                 | <i>2025-</i> |
| IEEE Transactions on Knowledge and Data Engineering (TKDE) | <i>2023-</i> |
| IEEE Transactions on Artificial Intelligence (TAI)         | <i>2023-</i> |